EMBL's European Bioinformatics Institute (EMBL-EBI)

PDBe-KB
Protein Data Bank in Europe - Knowledge Base

PDBe-KB: Comprehensive Clustering of Protein Conformations

Integrating Experimental and Computational Techniques

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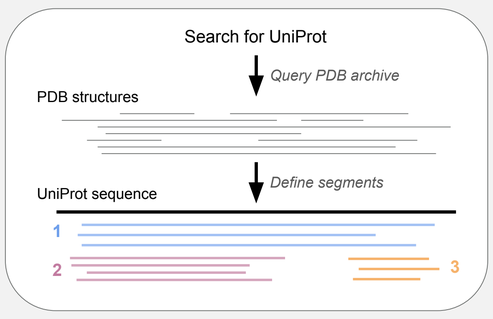
Introduction

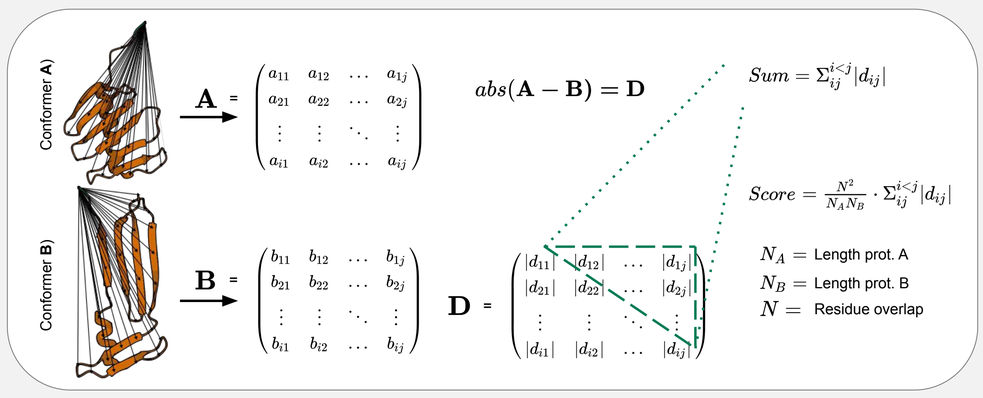
Proteins are dynamic macromolecules that can adopt specific conformations to carry out important biological functions. Understanding these shapes is essential for uncovering the molecular mechanisms driving biology. Advancements in AI-based protein structure prediction tools have further expanded the protein structure space. The Protein Data Bank (PDB) provides a sampling of protein conformers, but inconsistent annotations make it challenging to extract information on protein dynamics. The PDBe team has developed a standardised clustering pipeline that groups similar protein shapes together, enabling us to accurately identify distinct states or forms.

Methodology

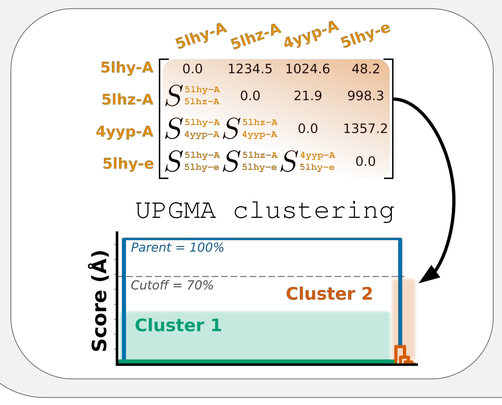
Step 1) Identify and group structures



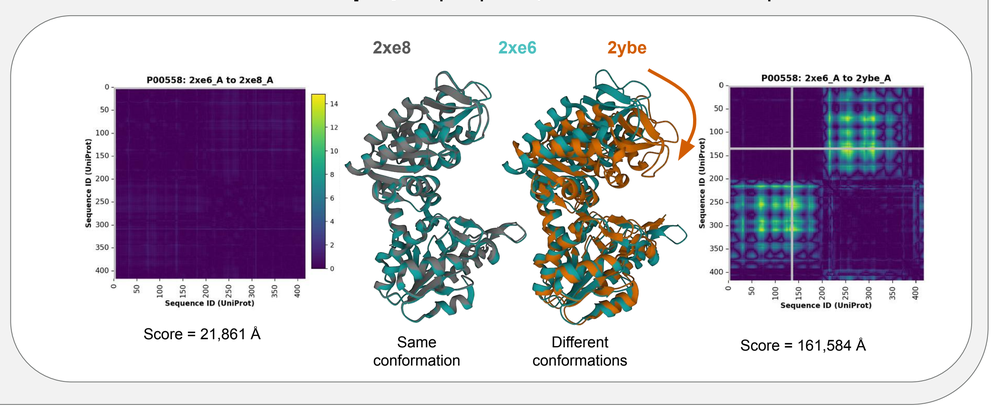
Step 2) Calculate "GLOCON" scores



Step 3) Cluster chains

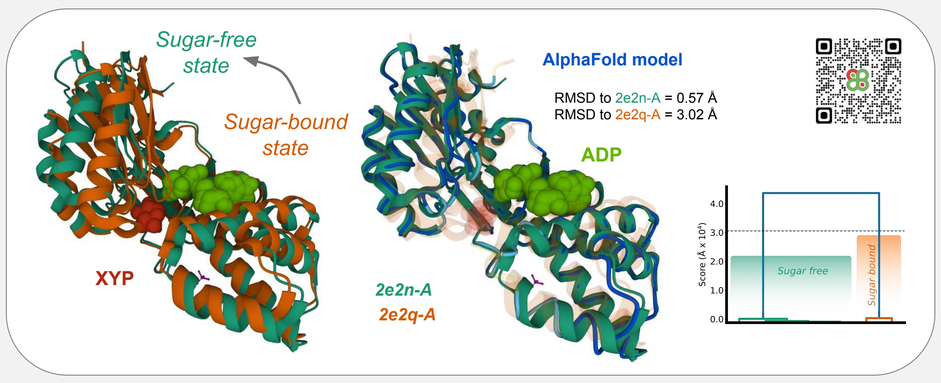


Step 4) Superpose, visualise and compare

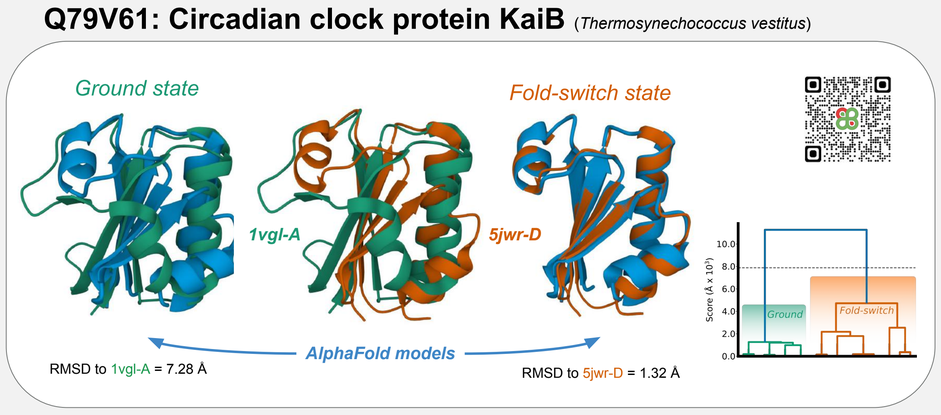


Gallery

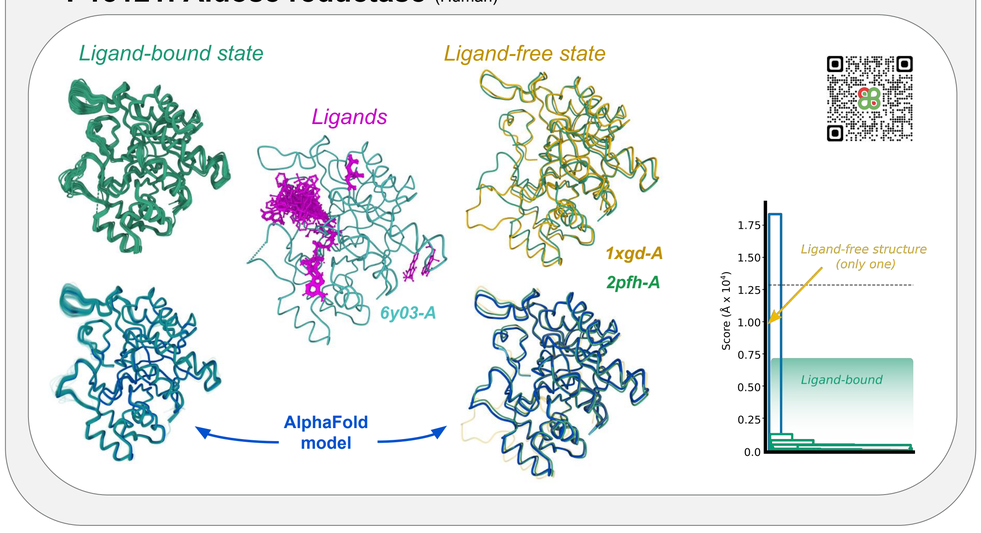
Q96Y14: Hexokinase (*Sulfurisphaera tokodaii*)



Q79V61: Circadian clock protein KaiB (*Thermosynechococcus vestitus*)

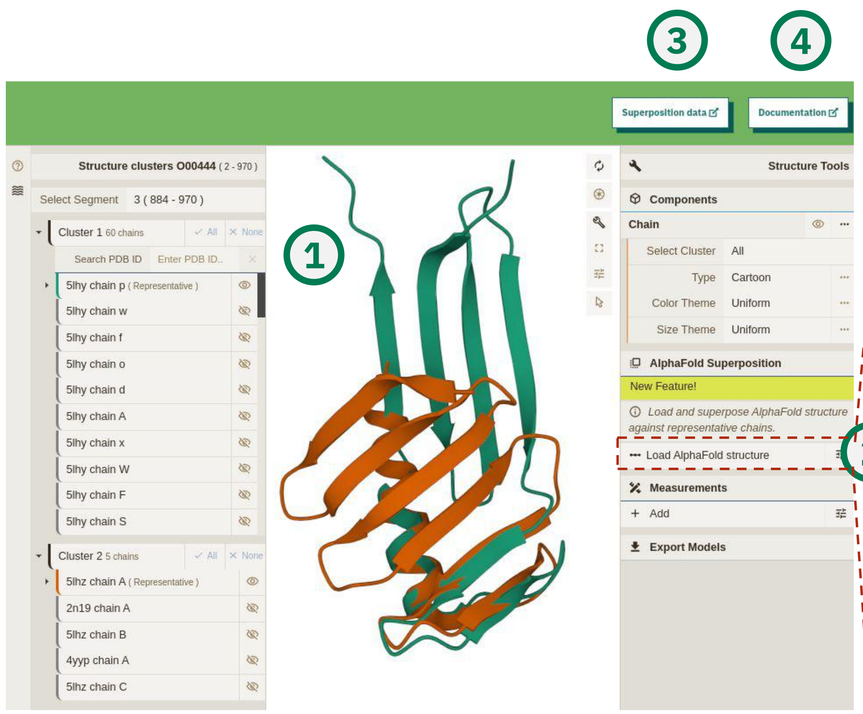


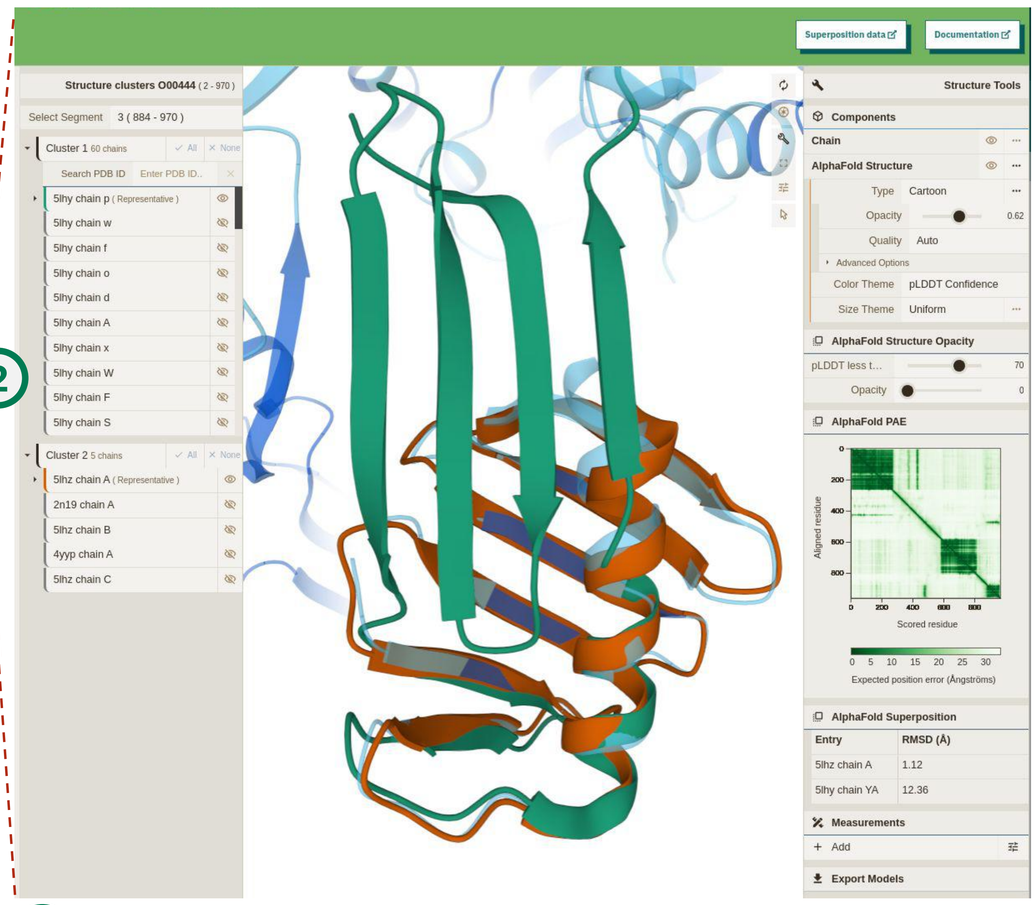
P15121: Aldose reductase (Human)

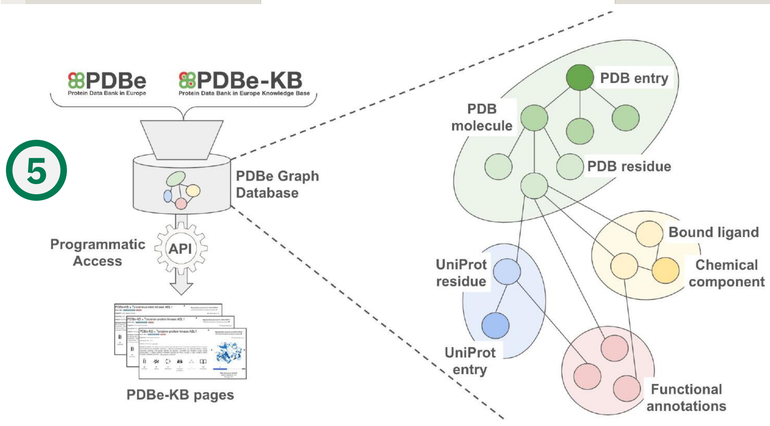


User Access

1. Superposed conformations can be visualised via the PDBe-KB website – click on the "3D view of superposed structures" button.
2. Superpose the predicted AlphaFold2 model with the "Load AlphaFold structure" button.
3. Download dendrograms and cluster annotation files.
4. Get more information on the pipeline by reading the documentation.
5. Use the GraphAPI to programmatically access recognised conformers.
6. Clone our open-source repository to contribute to the project or identify conformers on your own machine.







https://www.ebi.ac.uk/pdbe/graph-api/uniprot/superposition/<**uniprot_accession**>

6 Open-source code

bioRχiv
Pre-print coming soon...

Benchmark dataset

UK Research and Innovation
BBSRC bioscience for the future
CCP4
elixir

Acknowledgements
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